

# Zero Trust Access Policy with Okta Adaptive MFA

*Design and Implementation of Contextual Access Controls — documented by Rithik Kumar*

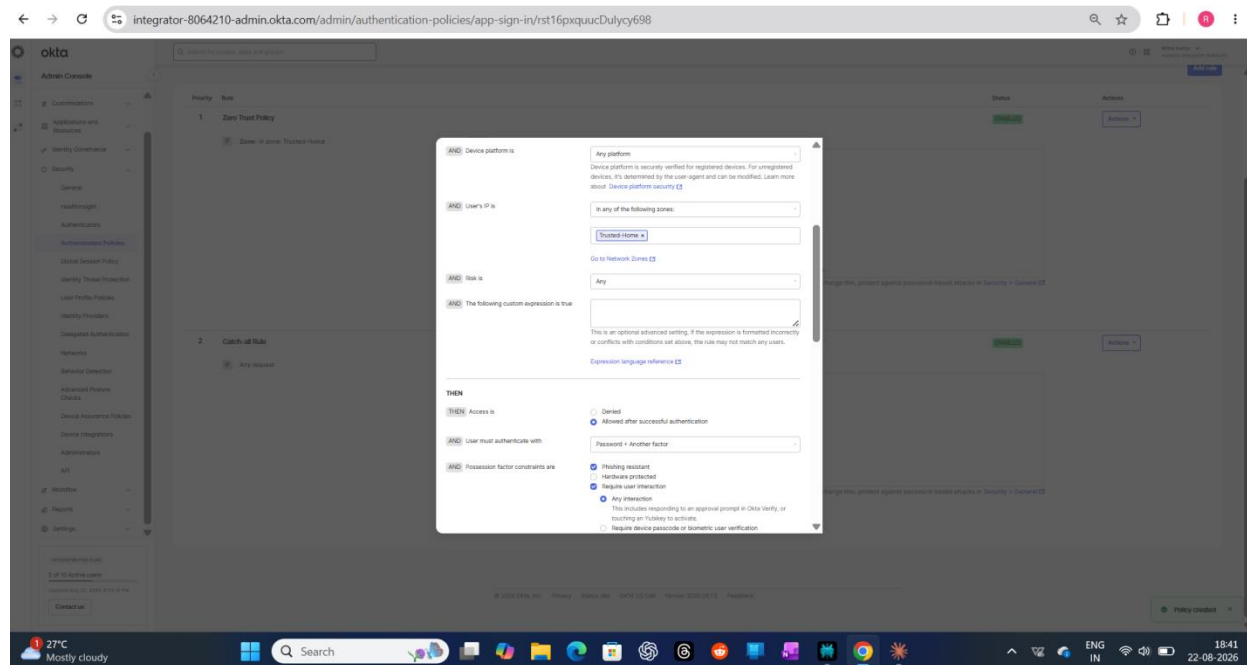
This project builds directly on the network zones and identity foundation from my earlier Okta work: a genuine adaptive authentication policy where the SAME login gets treated differently depending on context — which network it's coming from, what the device looks like, and what Okta's own risk signals say about it.

I'm including the parts of this that didn't go smoothly on purpose. I hit two separate errors while testing — a sign-in failure and a 403 Forbidden — and rather than just re-trying until it worked, I pulled the raw System Log data into a spreadsheet to actually find out what was happening underneath the UI. That debugging process is honestly the part of this project

## PHASE 1 — DESIGNING THE ADAPTIVE POLICY

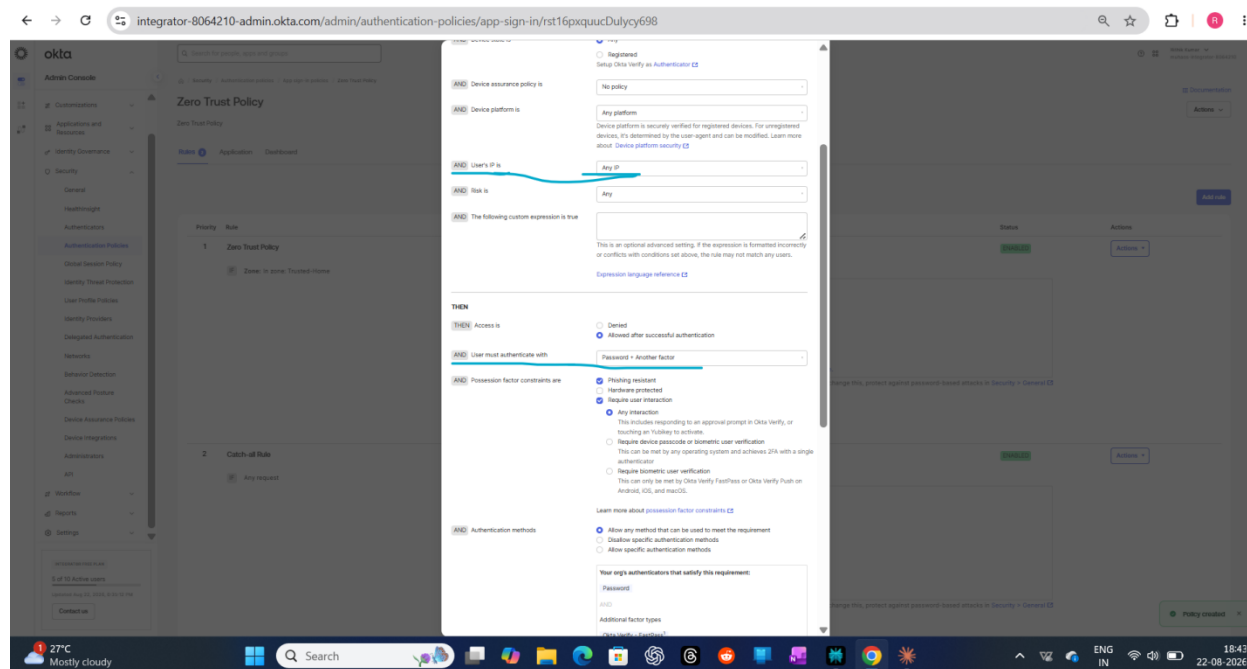
### Step 1 — Building the Rule Logic

Started by defining the actual conditions inside the policy rule — network zone, device platform, and risk level — and what should be required when they match: password plus another factor, with phishing-resistant and user-interaction constraints turned on.



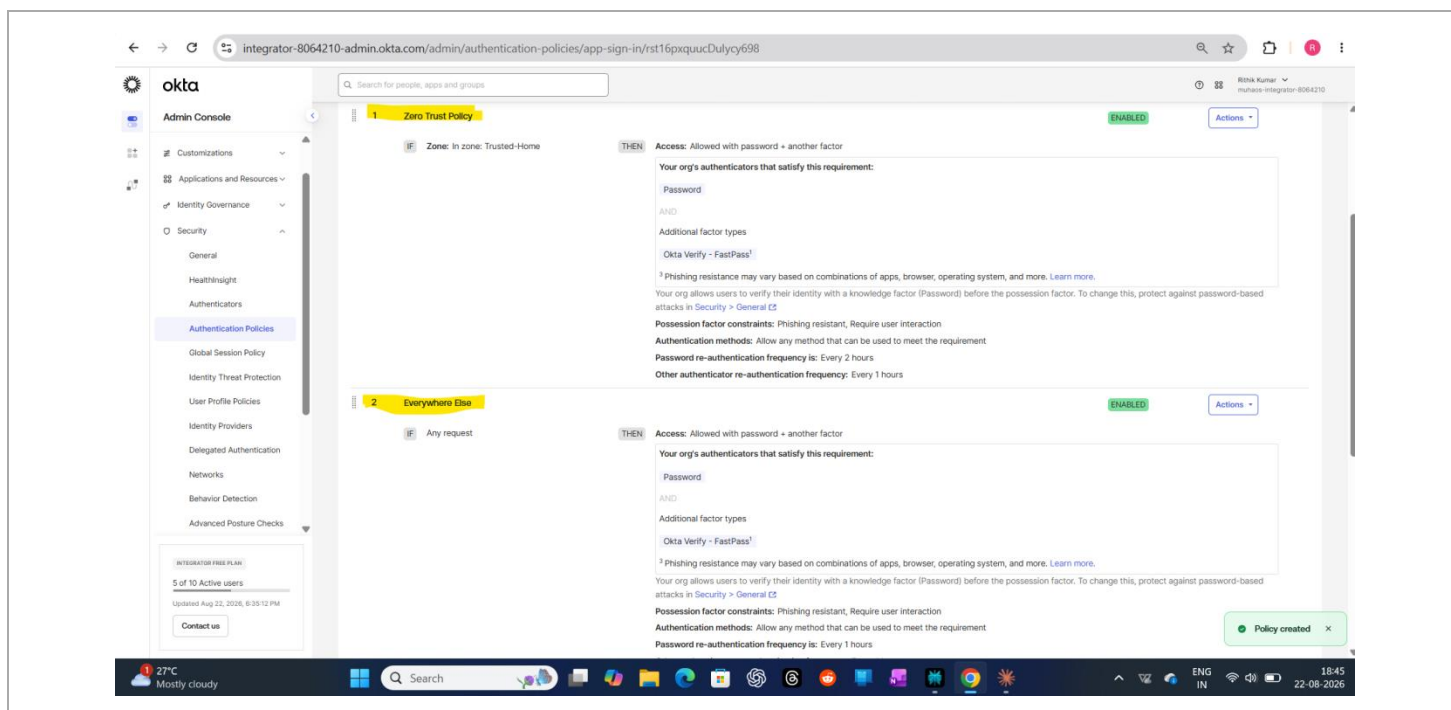
## Step 2 — Reviewing the Full Condition Set

Scrolled through the rest of the rule to double check the IP/zone condition and the authentication requirement together — wanted to actually understand every field before saving, not just click through it.



### Step 3 — Both Rules, Side by Side

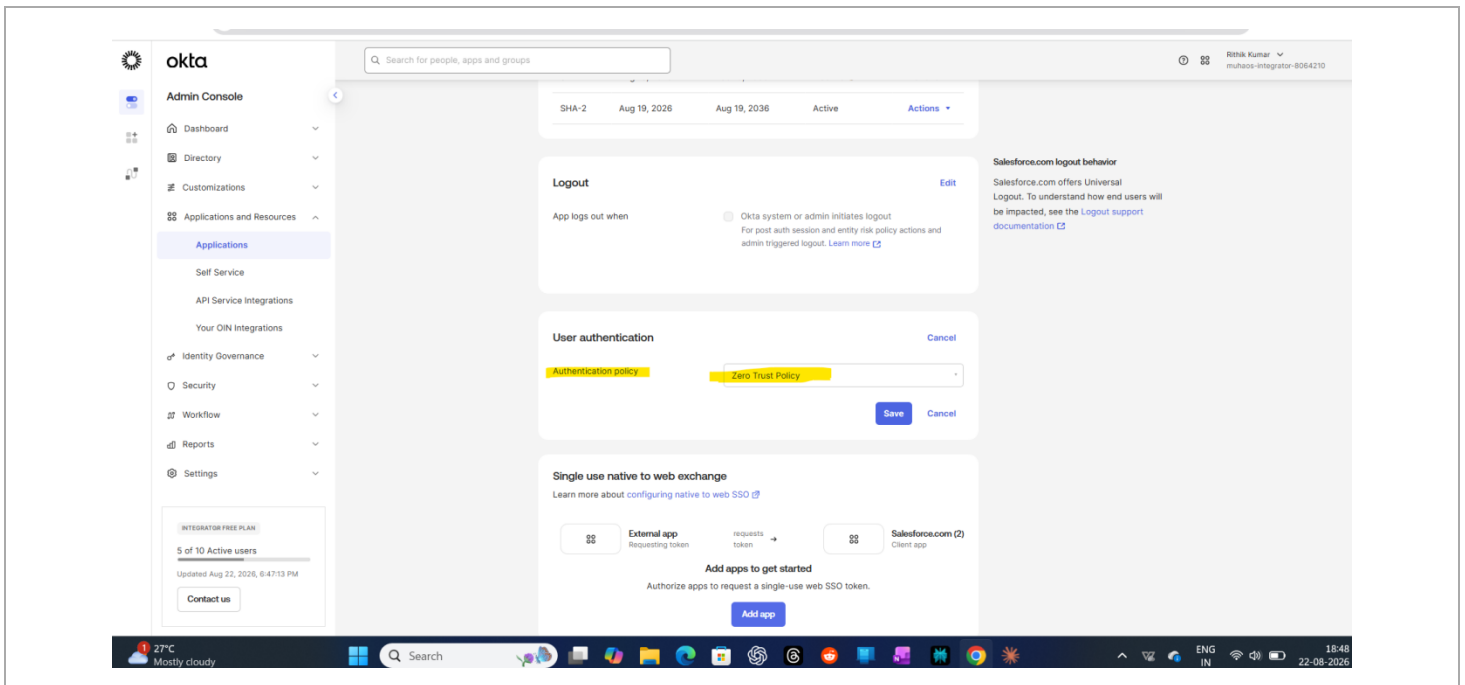
Final policy: Rule 1 matches the Trusted-Home zone, Rule 2 is the catch-all for everywhere else. I ended up requiring password + another factor on both rules rather than relaxing it for the trusted zone — in hindsight, a stricter baseline felt like the more defensible real-world choice than optimizing for convenience on my own home network.



## PHASE 2 — APPLYING IT TO A REAL APP

### Step 4 — Attaching the Policy to Salesforce

Switched the Salesforce app's authentication policy from Default to my new Zero Trust Policy — this is the point where the policy actually starts controlling real sign-ins, not just sitting there configured.



### PHASE 3 — A BONUS FIND

## Step 5 — Device Assurance Was Actually Available

Wasn't sure this would be available on the free tier, but it was — an existing Device Assurance policy checking screen lock, biometrics, disk encryption, root status, and Google Play Protect. Good reminder to just go check rather than assume a feature is locked behind a paid plan.

The screenshot displays the Okta Admin Console interface for a user named 'Bhishik Kumar' (mufassan-integrator-8064210). The left sidebar shows the 'Admin Console' menu with categories like Customizations, Applications and Resources, Identity Governance, and Security. The 'Security' section is expanded, showing options like General, HealthInsight, Authenticators, Authentication Policies, Global Session Policy, Identity Threat Protection, User Profile Policies, Identity Providers, Delegated Authentication, Networks, Behavior Detection, and Advanced Posture Checks. The 'Device Assurance Policies' page is active, featuring a table with one policy named 'Device Assurance policy' for the 'Android' platform. The policy conditions include: Screen lock must be enabled, Biometrics must be enabled, Device disk must be encrypted, Device must not be rooted, Google Play Protect must be enabled, and Scan threshold must be low risk. Below the table, the 'Remediation settings' section is visible, with options to link to general help or Okta documentation. The bottom of the screen shows a Windows taskbar with the date and time as 22-08-2026, 19:04.

Okta Admin Console

Search for people, apps and groups

Device Assurance Policies

Add a policy

| Policy name             | Platform | Conditions                                                                                                                                                                                                                     | Actions                 |
|-------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Device Assurance policy | Android  | Screen lock must be enabled<br>Biometrics must be enabled<br>Device disk must be encrypted<br>Device must not be rooted<br>Google Play Protect must be enabled<br>Scan threshold must be low risk<br><a href="#">Show less</a> | <a href="#">Actions</a> |

Remediation settings

When a user is denied access due to noncompliance with device assurance policies, the Sign-In Widget provides a link to remediation instructions.

Link to general help when access is denied or no information can be provided

☒ Link to Okta documentation

☐ Link to custom documentation of your choice

© 2026 Okta, Inc. Privacy Status site OK14 US Cell Version 2026.08.1 E Feedback

27°C Mostly cloudy

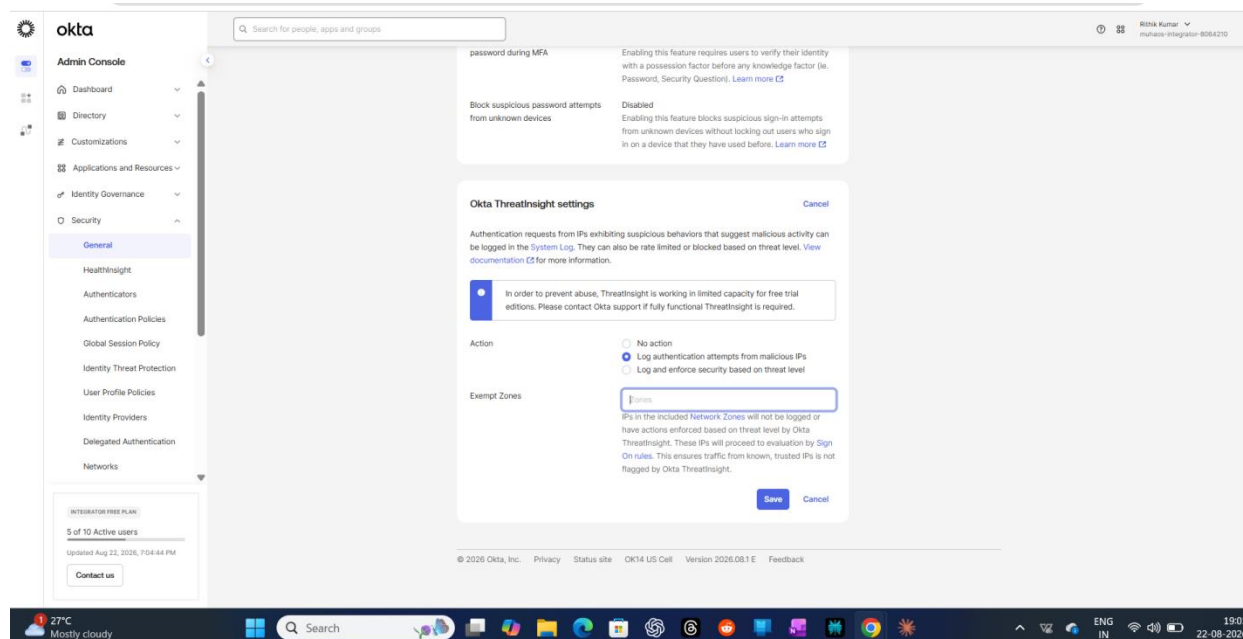
Search

ENG IN

19:04 22-08-2026

## Step 6 — Turning On ThreatInsight

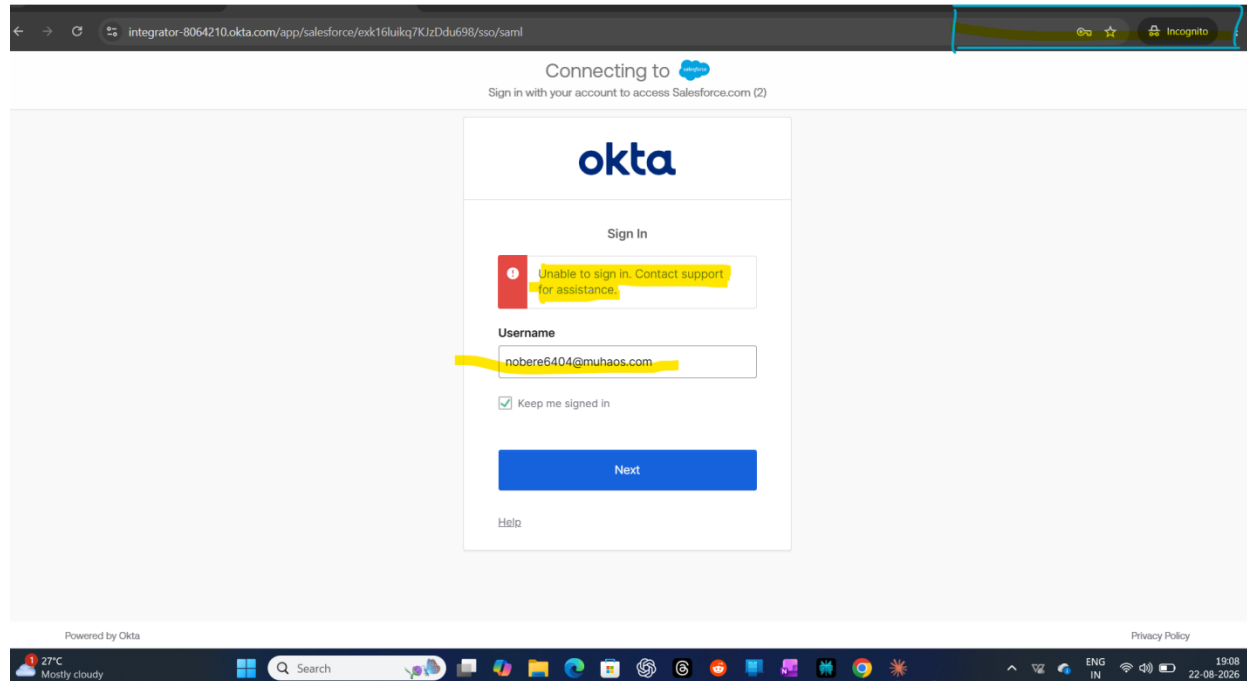
Enabled Okta ThreatInsight to log authentication attempts from suspicious IPs. It runs in limited capacity on the free tier, but it's still a real, functioning risk-signal layer on top of the zone-based rules.



## PHASE 4 — TESTING IT (AND HITTING REAL PROBLEMS)

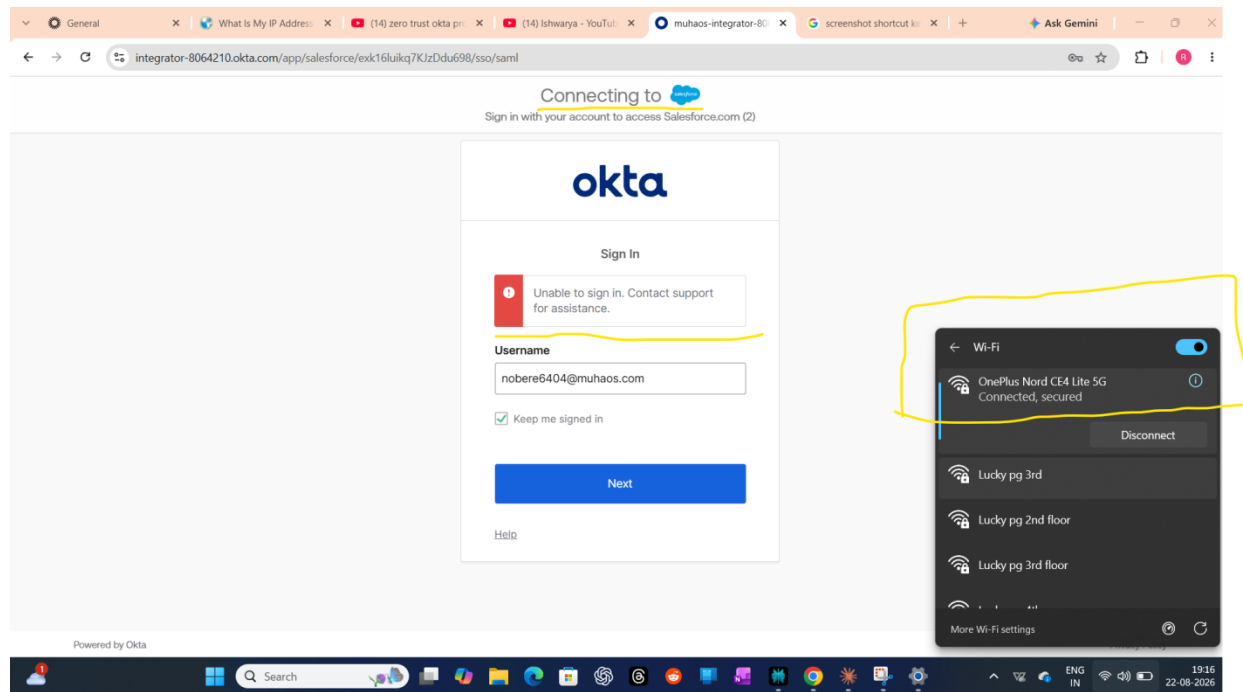
### Step 7 — First Real Test — An Error, Not a Clean Success

Tried signing in through an incognito window to test the policy cleanly. Got "Unable to sign in. Contact support for assistance." instead of the smooth pass/fail I expected. Honestly, my first reaction was to assume I'd broken the policy config.



## Step 8 — Same Error, Different Network

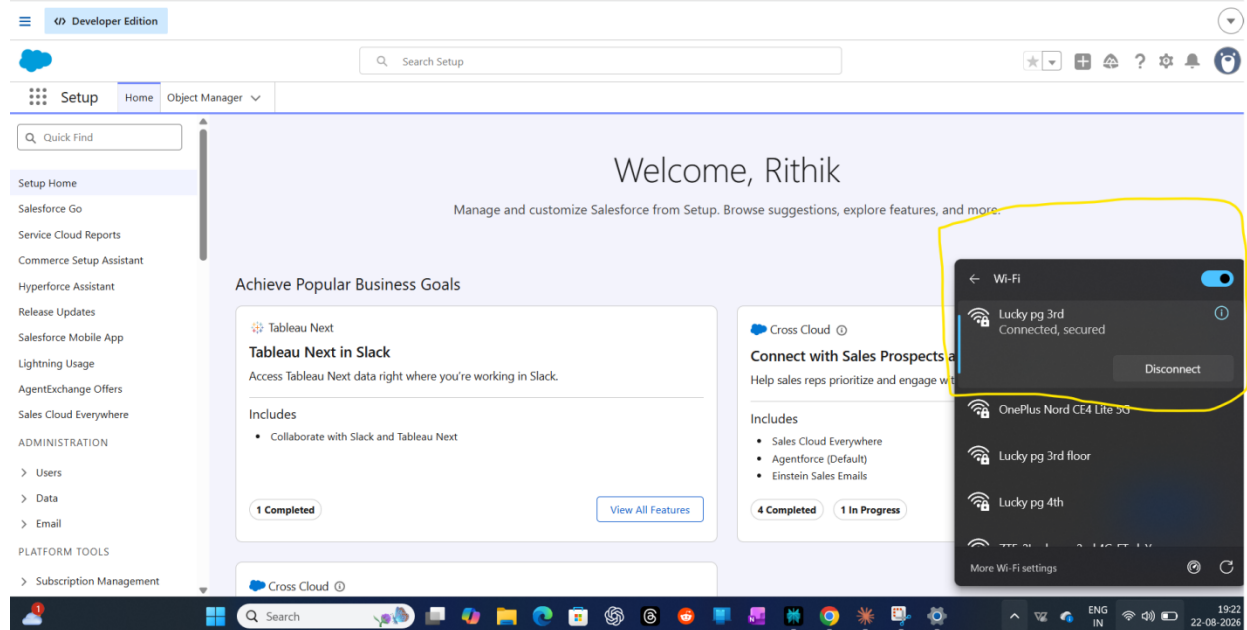
Switched to mobile network to properly test the "untrusted network" branch of the policy — same sign-in error came back. At this point I wasn't sure if it was the policy, the app config, or something else entirely.





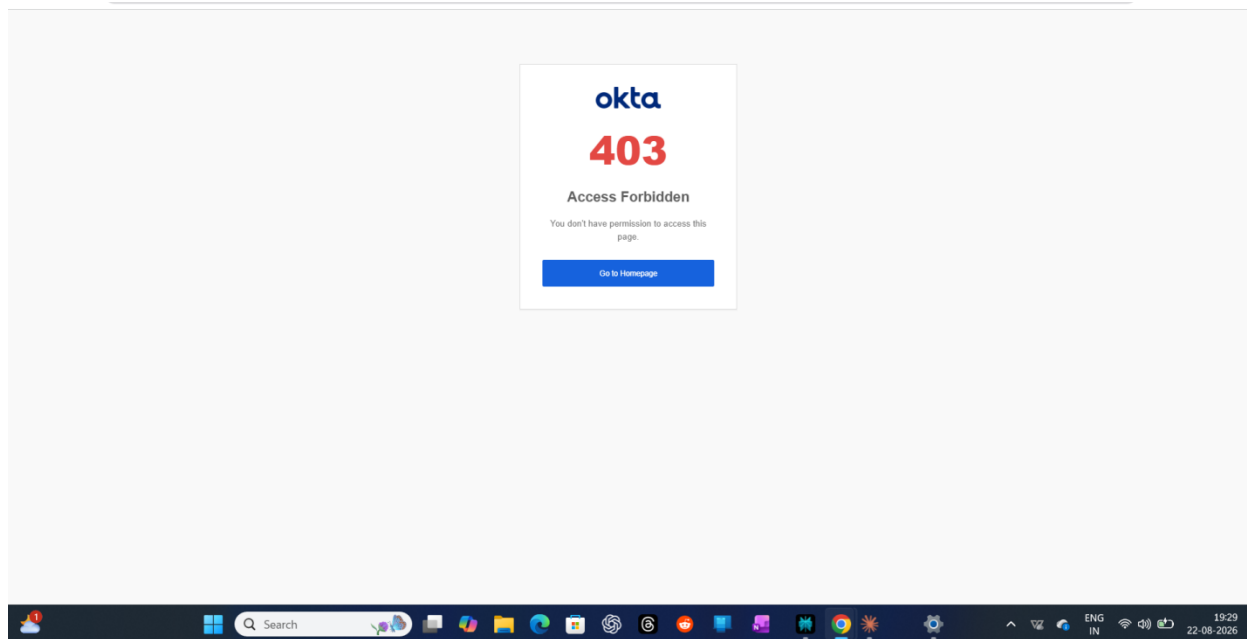
## Step 9 — Ruling Out the Network Itself

Cycled through a few different wifi networks to make sure the error wasn't just a flaky connection issue on my end before assuming it was a config problem on Okta's side.



## Step 10 — A Second, Different Error Shows Up

At one point hit a completely different error — a 403 Access Forbidden on the Okta side itself. Two different failure modes in one testing session was honestly a bit frustrating, but it meant there were two separate things to actually go investigate instead of one.



## Step 11 — Pulling the Raw System Log Into a Spreadsheet

You are currently in guest mode. Sign in to activate more features and services.

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[Upgrade to unlock unlimited PDF functionality, advanced AI tools, and 100,000+ high-quality templates.](#) [Upgrade Now](#)

| #   | L      | M | N                       | O | P | Q | R                         | S                    | T                             | U       | V                     | W      | X        | Y       | Z             | AA | AB | AC | AD              | AE | AF |
|-----|--------|---|-------------------------|---|---|---|---------------------------|----------------------|-------------------------------|---------|-----------------------|--------|----------|---------|---------------|----|----|----|-----------------|----|----|
| 136 | m@ok   | 0 |                         |   |   |   | trs_s3wgoJETHmswzS8huLadg |                      |                               |         |                       |        |          |         |               |    |    |    | i116nsr9foI JOB |    |    |
| 137 | m@ok   | 0 |                         |   |   |   | trs_s3wgoJETHmswzS8huLadg |                      |                               |         |                       |        |          |         |               |    |    |    | i116nsr9foI JOB |    |    |
| 138 | m@ok   | 0 |                         |   |   |   | trs_s3wgoJETHmswzS8huLadg |                      |                               |         |                       |        |          |         |               |    |    |    | i116nsr9foI JOB |    |    |
| 139 | m@ok   | 0 |                         |   |   |   | trs_s3wgoJETHmswzS8huLadg |                      |                               |         |                       |        |          |         |               |    |    |    | i116nsr9foI JOB |    |    |
| 140 | m@ok   | 0 |                         |   |   |   | trs_s3wgoJETHmswzS8huLadg |                      |                               |         |                       |        |          |         |               |    |    |    | i116nsr9foI JOB |    |    |
| 141 | m@ok   | 0 |                         |   |   |   | trs_s3wgoJETHmswzS8huLadg |                      |                               |         |                       |        |          |         |               |    |    |    | i116nsr9foI JOB |    |    |
| 142 | re6A04 | 0 | FACTOR_PR OKTA_CREC OTP |   |   |   | idx0d5dhj_l null          | 27.7.183.57 Computer | Mozilla/5.0 Windows 11 CHROME | India   | Bengaluru             | 562130 | 77.591   | 12.9753 | d8c3f1907 WEB |    |    |    | /fdp/dhc/c      |    |    |
| 143 | m@ok   | 0 |                         |   |   |   | trs_s3wgoJETHmswzS8huLadg |                      |                               |         |                       |        |          |         |               |    |    |    | i116nsr9foI JOB |    |    |
| 144 | re6A04 | 0 |                         |   |   |   | idx0d5dhj_l null          | 27.7.183.57 Computer | Mozilla/5.0 Windows 11 CHROME | India   | Bengaluru             | 562130 | 77.591   | 12.9753 | 940421200 WEB |    |    |    | /authn/p        |    |    |
| 145 | 000vhh | 0 |                         |   |   |   | idxlUIGD3c null           | 27.7.183.57 Computer | Mozilla/5.0 Windows 11 CHROME | India   | Bengaluru             | 562130 | 77.591   | 12.9753 | 2416c1e98 WEB |    |    |    | /login/error    |    |    |
| 146 | 000vhh | 0 |                         |   |   |   | idx0d5dhj_l null          | 44.238.82.1 Unknown  | OKta-Integr: Unknown          | UNKNOWN | United State Boardman | 97818  | -119.705 | 45.8401 | 0db33bc6c WEB |    |    |    | /oauth2/v1/y    |    |    |
| 147 | m@ok   | 0 |                         |   |   |   | trsrytTtHnRhOfnd1LygrA    |                      |                               |         |                       |        |          |         |               |    |    |    | i116m96i3g JOB  |    |    |
| 148 | m@ok   | 0 |                         |   |   |   | trsrytTtHnRhOfnd1LygrA    |                      |                               |         |                       |        |          |         |               |    |    |    | i116m96i3g JOB  |    |    |
| 149 | m@ok   | 0 |                         |   |   |   | trsrytTtHnRhOfnd1LygrA    |                      |                               |         |                       |        |          |         |               |    |    |    | i116m96i3g JOB  |    |    |
| 150 | m@ok   | 0 |                         |   |   |   | trsrytTtHnRhOfnd1LygrA    |                      |                               |         |                       |        |          |         |               |    |    |    | i116m96i3g JOB  |    |    |
| 151 | m@ok   | 0 |                         |   |   |   | trsrytTtHnRhOfnd1LygrA    |                      |                               |         |                       |        |          |         |               |    |    |    | i116m96i3g JOB  |    |    |
| 152 | m@ok   | 0 |                         |   |   |   | trsrytTtHnRhOfnd1LygrA    |                      |                               |         |                       |        |          |         |               |    |    |    | i116m96i3g JOB  |    |    |
| 153 | m@ok   | 0 |                         |   |   |   | trsrytTtHnRhOfnd1LygrA    |                      |                               |         |                       |        |          |         |               |    |    |    | i116m96i3g JOB  |    |    |
| 154 | m@ok   | 0 |                         |   |   |   | trslsIB8nbAQ5mj9YEmBIMl6w |                      |                               |         |                       |        |          |         |               |    |    |    | i116o5j2kh JOB  |    |    |
| 155 | m@ok   | 0 |                         |   |   |   | trslsIB8nbAQ5mj9YEmBIMl6w |                      |                               |         |                       |        |          |         |               |    |    |    | i116o5j2kh JOB  |    |    |
| 156 | m@ok   | 0 |                         |   |   |   | trslsIB8nbAQ5mj9YEmBIMl6w |                      |                               |         |                       |        |          |         |               |    |    |    | i116o5j2kh JOB  |    |    |
| 157 | m@ok   | 0 |                         |   |   |   | trslsIB8nbAQ5mj9YEmBIMl6w |                      |                               |         |                       |        |          |         |               |    |    |    | i116o5j2kh JOB  |    |    |
| 158 | m@ok   | 0 |                         |   |   |   | trslsIB8nbAQ5mj9YEmBIMl6w |                      |                               |         |                       |        |          |         |               |    |    |    | i116o5j2kh JOB  |    |    |
| 159 | m@ok   | 0 |                         |   |   |   | trslsIB8nbAQ5mj9YEmBIMl6w |                      |                               |         |                       |        |          |         |               |    |    |    |                 |    |    |

## Step 12 — Confirmed: The Policy Genuinely Works

Went back into the System Log UI and found it — a successful "Verify user identity" event through Okta Verify, tied to my account, from my actual location. 1,753 events logged across all my testing. The adaptive MFA challenge was firing correctly; the earlier errors turned out to be session/browser-state issues from switching networks mid-session, not a broken policy.

The screenshot displays the Okta Admin Console interface. The left sidebar shows the 'Admin Console' menu with options like Dashboard, Directory, Customizations, Applications and Resources, Identity Governance, Security, Workflow, Reports, and System Log. The 'System Log' section is active, showing a search bar and a 'Count of events over time' bar chart. Below the chart, a table lists events. A specific event is highlighted, showing details for a user named 'Rithik Kumar' who successfully verified their identity. The event details include the actor's email, display name, ID, type, client device, and geographical context. A 'Download CSV' button is visible next to the event list.

Okta Admin Console

System Log

Events: 1753

Download CSV

| Time            | Actor                                  | Event Info                      | Targets                                                               |
|-----------------|----------------------------------------|---------------------------------|-----------------------------------------------------------------------|
| Aug 22 19:31:09 | Rithik Kumar (User)<br>106.192.228.107 | Verify user identity<br>SUCCESS | Okta Verify (AuthenticatorMethod)<br>Okta Admin Console (AppInstance) |

Actor Details:

- AlternateId: nobers6404@muhaos.com
- DetailEntry: Rithik Kumar
- DisplayName: Rithik Kumar
- ID: 00u100wh6buYuroL898
- Type: User
- Client: Computer
- Device: Computer
- GeographicalContext: Bengaluru
- Country / region: India

## **A Personal Note on This Project (and the Three Before It)**

This is the third hands-on project I've built this month, and I want to be honest about something: before this, my IAM knowledge was mostly certifications, study material, and watching other people configure things in videos. Reading about SAML, MFA policies, and Zero Trust concepts is one kind of understanding. Actually sitting with a broken sign-in error at 7 PM, not knowing if it's my policy, my network, or the app config — and working through it step by step until the System Log finally showed me the truth — is a completely different kind of understanding.

If you're hiring for an IAM Engineer role, or know someone who is — I'd love to talk.